Automated Spectrum Peak Extraction

THE THREE ALGORITHMS

1

Greedy Approach

Matches the peaks and labels locally.

2

Dynamic Programming

Matches the peak and labels in a sequence to give a monotonic solution

3

Graph Based

Performs 1-1 matching using bipartite graphs

METRICS

- Precision: Proportion of predicted peaks that are correct.

$$\text{Precision} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Positives (FP)}}$$

- Recall: Proportion of true (ground-truth) peaks that are detected.

$$\text{Recall} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}}$$

- F1 Score: Harmonic mean of Precision and Recall.

$$\text{F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

- True Positives,False Positives and False Negatives (TP/FP/FN): Counts of correct matches, extra predictions, and missed ground-truths.

- RMSE (Relative Abundance): Root Mean Square Error between predicted and actual peak intensities. Lower is better. Range : 0-100

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_{i,\text{pred}} - y_{i,\text{true}})^2}$$

- Levenshtein Distance (Name Match): Number of single-character changes needed to turn one name into another. 0 = perfect match; higher = more differences.

GREEDY APPROACH

- Loads each PDF in the folder.
- Detects x-axis, y-axis, key plot segments, and the plot box region.
- Extracts the chemical name using regex.
- Identifies all candidate numbers within the plot box as spectral labels.
- Assigns each number to its closest peak (using horizontal and vertical proximity).
- Computes relative abundance for each peak based on vertical height from the baseline (with respect to the "100" tick).

METRIC	AVG VALUE (FOR 12 FILES)
Precision (For peak numbers)	1.0
Recall (For peak numbers)	1.0
F1 score (For peak numbers)	1.0
True Positives	219
False Positives	0
False Negatives	0

METRIC	AVG VALUE (FOR 12 FILES)
RMSE (for Relative Abundance)	0.3009558690714648
Name matching precision (Leveshtein Distance)	0.0

METRIC	VALUE
Total Time (for 25 files)	3.368s
Time for Core logic (for 25 files)	0.014s

RMSE CALCULATION EXAMPLE FOR FILE 74002.PDF

Ground Truth : $y_{i\ pred}$	Prediction : $y_{i\ true}$	$y_{i\ pred} - y_{i\ true}$	$(y_{i\ pred} - y_{i\ true})^2$
2.356020942	2.438879933	-0.082859	0.00687
7.068062827	7.142430135	-0.074367	0.005532
3.926701571	3.832512952	0.094189	0.008872
100	99.99405235	0.005948	0.000035
5.497382199	5.443927434	0.053455	0.00286
5.759162304	5.705236394	0.053926	0.002911
0.2617801047	0	0.26178	0.068529
4.97382199	4.921298438	0.052524	0.002758
0	0	0	0
0.2617801047	0.217748237	0.044032	0.001939
0.5235602094	0.435507549	0.088053	0.00775
1.308900524	1.437193741	-0.128293	0.016468
0.7853403141	0.740377232	0.044963	0.002023
0.2617801047	0.435507549	-0.173727	0.030205
1.047120419	1.132335134	-0.085215	0.007264
0	0.130648942	-0.130649	0.01708
0.2617801047	0.304858607	-0.043079	0.001857
0	0.130648942	-0.130649	0.01708
3.664921466	3.52766542	0.137256	0.018846
			Sum = 0.218879

$$\text{RMSE} = \sqrt{\frac{0.218879}{19}} = 0.1073$$

GT peaks=[42, 56, 70, 100, 115, 141, 152, 169, 181, 200, 226, 254, 268, 283, 298, 312, 353, 368, 398],

Pred peaks=[42, 56, 70, 100, 115, 141, 152, 169, 181, 200, 226, 254, 268, 283, 298, 312, 353, 368, 398]

GT chemical name : JWH 193

Pred chemical name: JWH 193

DYNAMIC PROGRAMMING

- Loads PDFs from the folder.
- Detects axes, plot box, and all vertical segments (candidate peaks).
- Extracts chemical name.
- Identifies all peaks by analyzing vertical lines anchored at the baseline.
- For each peak, searches for the closest numeric label using horizontal proximity within the plot box.
- Assigns labels to peaks using a dynamic programming approach to maintain optimal left-to-right assignment.
- Calculates relative abundance for each matched label, normalized against the y-axis (100 tick).

METRIC	AVG VALUE (FOR 12 FILES)
Precision (For peak numbers)	1.0
Recall (For peak numbers)	1.0
F1 score (For peak numbers)	1.0
True Positives	219
False Positives	0
False Negatives	0

METRIC	AVG VALUE (FOR 12 FILES)
RMSE (for Relative Abundance)	1.29341019120492
Name matching precision (Leveshtein Distance)	0.0

METRIC	VALUE
Total Time (for 25 files)	0.571s
Time for Core logic (for 25 files)	0.022s

RMSE CALCULATION EXAMPLE FOR FILE 74002.PDF

Ground Truth : $y_{i\ pred}$	Prediction : $y_{i\ true}$	$y_{i\ pred} - y_{i\ true}$	$(y_{i\ pred} - y_{i\ true})^2$
2.356020942	2.438879933	-0.082858991	0.006867
7.068062827	7.142430135	-0.074367308	0.005532
3.926701571	3.832512952	0.094188619	0.008873
100	99.99405235	0.00594765	0.000035
5.497382199	5.443927434	0.053454765	0.00286
5.759162304	2.134010251	3.625152053	13.147787
0.261780105	5.705236394	-5.443456288	29.631195
4.97382199	4.921298438	0.052523552	0.002757
0	0.914575822	-0.914575822	0.836426
0.261780105	0.217748237	0.044031868	0.001938
0.5235602094	0.435507549	0.088052661	0.007753
1.308900524	1.437193741	-0.128293217	0.016467
0.7853403141	0.740377232	0.044963082	0.002023
0.261780105	0.435507549	-0.173727444	0.030192
1.047120419	1.132335134	-0.085214715	0.007265
0	0.304858607	-0.304858607	0.092943
0.261780105	0.304858607	-0.043078502	0.001857
0	0.304858607	-0.304858607	0.092943
3.664921466	3.52766542	0.137256046	0.018845
			Sum : 43.914558

$$\text{RMSE} = \sqrt{\frac{43.914558}{19}} = 1.5202$$

GT peaks=[42, 56, 70, 100, 115, 141, 152, 169, 181, 200, 226, 254, 268, 283, 298, 312, 353, 368, 398],

Pred peaks=[42, 56, 70, 100, 115, 141, 152, 169, 181, 200, 226, 254, 268, 283, 298, 312, 353, 368, 398]

GT chemical name : JWH 193

Pred chemical name: JWH 193

GRAPH BASED

- Loads PDFs from the folder.
- Detects plot axes, segments, and candidate plot box.
- Extracts chemical name.
- Collects all vertical segments (peaks) and numeric labels (candidate spectrum points).
- Constructs a cost matrix between all possible label–peak pairs, based on proximity and alignment. (cost function)
- Uses the Hungarian algorithm (graph matching) to optimally assign each label to a peak, minimizing total assignment cost.
- Computes relative abundance using peak heights relative to the baseline and 100 tick.

METRIC	AVG VALUE (FOR 12 FILES)
Precision (For peak numbers)	1.0
Recall (For peak numbers)	1.0
F1 score (For peak numbers)	1.0
True Positives	219
False Positives	0
False Negatives	0

METRIC	AVG VALUE (FOR 12 FILES)
RMSE (for Relative Abundance)	0.295322180396116
Name matching precision (Leveshtein Distance)	0.0

METRIC	VALUE
Total Time (for 25 files)	0.527s
Time for Core logic (for 25 files)	0.016s

RMSE CALCULATION EXAMPLE FOR FILE 74002.PDF

Ground Truth : $y_{i\ pred}$	Prediction : $y_{i\ true}$	$y_{i\ pred} - y_{i\ true}$	$(y_{i\ pred} - y_{i\ true})^2$
2.356020942	2.438879933	-0.082859	0.00687
7.068062827	7.142430135	-0.074367	0.005532
3.926701571	3.832512952	0.094189	0.008872
100	99.99405235	0.005948	0.000035
5.497382199	5.443927434	0.053455	0.00286
5.759162304	5.705236394	0.053926	0.002911
0.261780105	0.217748237	0.044032	0.001939
4.97382199	4.921298438	0.052524	0.002758
0	0	0	0
0.261780105	0.217748237	0.044032	0.001939
0.5235602094	0.435507549	0.088053	0.00775
1.308900524	1.437193741	-0.128293	0.016468
0.7853403141	0.740377232	0.044963	0.002023
0.261780105	0.435507549	-0.173727	0.030201
1.047120419	1.132335134	-0.085215	0.007264
0	0	0	0
0.261780105	0.304858607	-0.043079	0.001857
0	0	0	0
3.664921466	3.52766542	0.137256	0.018846
			Sum = 0.118125

$$\text{RMSE} = \sqrt{\frac{0.118125}{19}} = 0.0788$$

GT peaks=[42, 56, 70, 100, 115, 141, 152, 169, 181, 200, 226, 254, 268, 283 298, 312, 353, 368, 398],

Pred peaks=[42, 56, 70, 100, 115, 141, 152, 169, 181, 200, 226, 254, 268, 283, 298, 312, 353, 368, 398]

GT chemical name : JWH 193

Pred chemical name: JWH 193

THANK YOU